GGG-P-836
DECEMBER 8, 1954
SUPERSEDING
Int. Fed. Spec. GGG-P-00836 (NAVY-SHIPS)
20 January 1954

FEDERAL SPECIFICATION

PUNCH AND DIE, KNOCKOUT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers light- and heavy-duty punches and dies used for cutting smooth holes in plastic, hard rubber, and metal.

1.1.1 *Federal specification coverage.*—This specification covers only the types, classes, designs, and styles of the commodity generally purchased by the Federal Government, and does not include all types, classes, designs, and styles which are commercially available.

1.2 Classification.

1.2.1 *Types and classes.*—The punches shall be of the following types and classes, as specified. (See 6.1.)

Type I.—Light duty (chassis).
Class 1.—Round hole.
Class 2.—Square hole.
Class 3.—Keyed hole.
Type II.—Heavy duty (conduit).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications.—There are no specifications applicable to this specification other than those referenced in section 7 by individual departments.

2.2 Standards.—The following Military standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

2.3 Other publications.—The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

Governmental:

National Bureau of Standards Publication:
Handbook H28—Screw Thread Standards for Federal Services.

Nongovernmental:

Consolidated Classification Committee Publication:
Consolidated Freight Classification Rules.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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(Application for copies of National Bureau of Standards Handbook H28 should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price may be obtained from the Superintendent of Documents.

(Application for copies of Consolidated Freight Classification Rules should be addressed to the Consolidated *Classification Committee, 202 Chicago Union Station*, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material.—The material employed in the construction of the punches and dies shall be free from all defects and imperfections that may affect the serviceability or durability. When *materials are used which are not specifically designated*, they shall be entirely suitable for the purpose intended.

3.1.1 *Punches, dies, and drivescrews.*—The punches, dies, and drivescrews, shall be of a high-quality alloy steel.

3.1.2 *Nuts.*—The nuts shall be of a good-quality steel.

3.2 Screw threads.—Screw threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.3 Hardness.

3.3.1 *Punch.*—The punch shall be hardened and tempered showing a hardness of not less than 55 nor more than 60 on a Rockwell C scale.

3.3.2 *Die.*—The die shall be hardened and tempered showing a hardness of not less than 55 nor more than 58 on a Rockwell C scale or be casehardened to a minimum depth of 0.030 inch showing a hardness of not less than 87 nor more than 90 on a Rockwell superficial 15-N scale, under 15-kilogram load.

3.4 Interchangeability of parts.—All parts having the same manufacturer's part number shall be constructed to definite standards, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.4.1 *Illustrations.*—The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of punches otherwise conforming to this specification.

3.5 Marking for identification.—The punches and dies shall be marked in a plain and permanent manner with the size and the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.6 Type I, light duty (chassis).—Type I punches shall consist essentially of a punch, die, and drivescrew, or drivescrew and nut. The punches shall be capable of cutting clean accurate holes in plastic, hard rubber, and metal equal to 14-gage (0.075-inch) mild sheet steel.

3.6.1 *Punch.*—The punch shall have the outer surfaces smoothly machined or ground. The drive-end bearing surface shall be at a right angle to the axis of the punch. The cutting end of the punch shall be machined or ground to cut with a shearing action.

3.6.2 *Die.*—The die shall have the top and bottom surfaces smoothly machined or ground at a right angle to the axis of the die. The surface of the hole in the die accommodating the punch shall be smoothly machined or ground.

3.6.3 *Drivescrew.*—The drivescrew shall be hardened and tempered showing not less than 45 nor more than 50 on a Rockwell C scale.

3.6.4 *Drive nut.*—The drive nut shall be either square or hexagon in shape having a threaded center hole to engage the thread on the drivescrew. The driving surface of the nut shall be smoothly machined at right angle to the axis of the threaded center hole of the drive nut. The drive nut shall be suitably casehardened.

3.6.5 *Class 1, round hole.*—Class 1 punches shall be designed to cut round holes accomplishing the punching operation by the use of a drivescrew.

3.6.5.1 *Punch.*—The punch shall have a threaded center hole to accommodate the thread of the drivescrew. The outside diameter of the punch shall be concentric with the threaded center hole.

3.6.5.2 *Die.*—The die shall be provided with a smoothly machined center hole to accommodate the body of the drivescrew. The hole ac-

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commodating the punch shall be of a suitable depth and concentric with the center hole.

3.6.5.3 *Drivescrew.*—The drivescrew shall have either a square or hexagon-shaped head. The surface at the head of the screw which bears against the die shall be flanged. The driving surface of the screw shall be smoothly machined at right angles to the axis of the body of the screw.

3.6.5.4 Class 1 punches shall be capable of punching holes conforming to the sizes shown in table I, and shall be similar to figure 1. (See 6.1.)

3.6.6 *Class 2, square hole.*—Class 2 punches shall be designed to cut square holes, accomplishing the punching action by the use of a drivescrew and nut.

3.6.6.1 *Punch.*—The punch shall be provided with a suitably shaped center hole to accomplish self-alignment of the punch when assembled to the drivescrew.

3.6.6.2 *Die.*—The outer surface of the die shall be either circular or square in shape. The die shall have a suitable squared hole to accommodate the punch and suitable means shall be provided to neatly fit on the drivescrew to insure proper alignment with the punch when assembled to the drivescrew.

3.6.6.3 *Drivescrew.*—The drivescrew shall have either a square or hexagon-shaped head and shall be threaded on the opposite end to engage the drive nut to accomplish the punching action. The surfaces of the drivescrew which aligns the punch and die shall be machined to close tolerances insuring proper alignment of the punch with the die.

3.6.6.4 Class 2 punches shall be capable of punching holes conforming to the sizes shown in table II, and shall be similar to figure 2 (see 6.1).

TABLE I.—*Type I, class 1, light duty (chassis), round hole*

Hole size	Drivescrew size	Drivescrew flange (min.)
Inches	*Inch*	*Inch*
½	¼	25/64
9/16	¼	25/64
11/16	5/16	5/8
¾	5/16	5/8
13/16	5/16	5/8
1	5/16	5/8
1 1/16	5/16	5/8
1 1/8	5/16	5/8
1 3/16	5/16	5/8
1 11/64	5/16	5/8
1 5/16	5/16	5/8
1 ¼	5/16	5/8
1 3/8	5/16	5/8
1 ½	5/16	5/8
1 5/8	¾	—
1 ¾	¾	—
1 7/8	¾	—
2 ¼	—	—

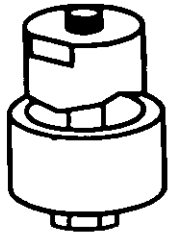


FIGURE 1.—Type I, class 1, light duty (chassis), round hole.

TABLE II.—*Type I, class 2, light duty (chassis), square hole*

Hole size
Inch
5/8
11/16
¾
7/8
1

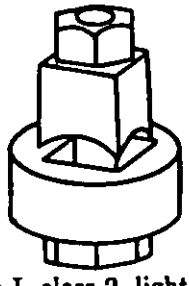


FIGURE 2.—Type I, class 2, light duty (chassis), square hole.

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3.6.7 *Class 3, keyed hole.*—Class 3 punches shall be designed to cut keyed holes, accomplishing the punching action by the use of the drivescrew and nut.

3.6.7.1 *Punch.*—The punch shall be circular in shape having a keyway on the outer diameter suitably positioned to engage the key in the die. The punch shall be provided with a suitably shaped center hole to accomplish self-alignment of the punch when assembled to the drivescrew.

3.6.7.2 *Die.*—The body of the die shall be circular in shape having a circular hole with a key to engage the keyway in the punch.

3.6.7.3 *Drivescrew.*—The drivescrew shall be in accordance with 3.6.6.3.

3.6.7.4 Class 3 keyed-hole punches shall be capable of punching holes conforming to the sizes shown in table III, and shall be similar to figure 3 (see 6.1).

TABLE III.—*Type I, class 3, light duty (chassis), keyed hole*

Hole size A	Key size		
	B	C	D
+0.010 − .000	Min.	Min.	Min.
Inches	*Inch*	*Inch*	*Inch*
15/16	0.147	0.147	0.093
1 11/64	.155	.110	.087
1 17/64	.147	.147	.093
1 21/64	.146	.093	.112

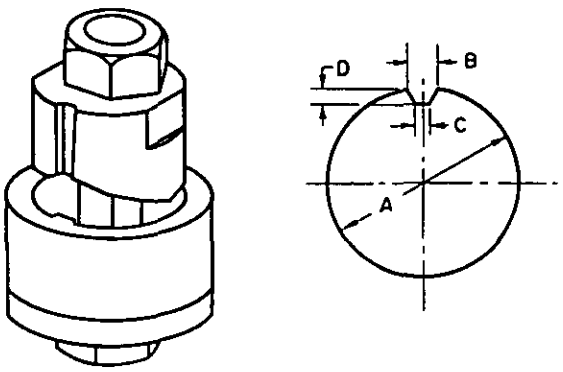


FIGURE 3.—Type I, class 3, light duty (chassis), keyed hole.

3.7 Type II, heavy duty (conduit).—Type II punches shall be designed to cut round holes, accomplishing the punching action by means of a drivescrew or drivescrew and nut. The punch shall consist essentially of a punch, die, and drivescrew, or drivescrew and nut. The punches shall be capable of cutting clean accurate holes in plastic, hard rubber, and metal equal to 10-gage (0.135-inch), mild, sheet steel.

3.7.1 *Punch.*—The punch shall be in accordance with 3.6.1 and 3.6.5.1.

3.7.2 *Die.*—The die shall be in accordance with 3.6.2 and 3.6.5.2.

3.7.3 *Drivescrew.*—The drivescrew shall have either a square or hexagon-shaped head. The driving surface of the screwhead shall be smoothly machined at a right angle to the axis of the body of the drivescrew. The body of the drivescrew shall be machined to close tolerances insuring proper alignment of the punch with the die and to engage the thread of the drive nut to effect the punching action. The drivescrew shall be suitably heat-treated to possess a minimum tensile strength of 200,000 p. s. i.

3.7.4 *Drive nut.*—The drive nut shall be in accordance with 3.6.4.

3.7.5 Type II punches shall be capable of punching round holes conforming to the sizes shown in table IV, and shall be similar to figure 4 or 5.

TABLE IV.—*Type II, heavy duty (conduit)*

Hole size	Conduit size
Inches	*Inches*
7/8	½
1 3/32	¾
1 11/32	1
1 11/16	1¼
1 15/16	1½
2 3/8	2
2 7/8	2½
3½	3
4	3½
4½	4

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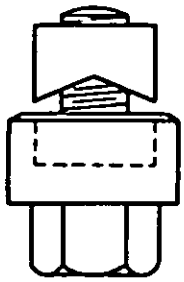


FIGURE 4.—Type II, heavy duty (conduit).

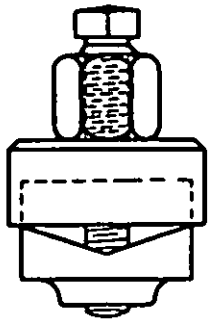


FIGURE 5.—Type II, heavy duty (conduit).

3.8 Workmanship.—Workmanship shall be of the highest grade throughout and equal in every respect to the best commercial practice. The punches and dies shall be free from rust, fins, burrs and any other imperfection which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All punches and dies of the same type, class and size presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

4.1.2 *Sampling for inspection.*—A random sample of punches and dies shall be taken by the Government inspector from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements of table V in accordance with Military Standard MIL–STD–105.

TABLE V.—*Sampling for visual and dimensional inspection A. Q. L. (approx.) equals 1.5 percent defective*

Number of punches and dies in the inspection lot	Number of punches in the sample
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with Section 9 of Military Standard MIL–STD–105.

4.1.3 *Sampling for lot acceptance tests.*

4.1.3.1 *Sampling for hardness test.*—A random sample of punches and dies shall be taken by the Government inspector from each inspection lot in accordance with table V and shall be subjected to the hardness tests specified in 4.4.1.

4.1.3.2 *Sampling for punch-cutting test.*—The sample for punch-cutting tests shall be one-fifth of the sample of 4.1.2 (single sampling only).

4.1.3.3 *Sampling for tensile strength of drivescrews.*—Two drivescrews shall be removed from punches which have been inspected, and these screws shall be subjected to the tensile tests specified in 4.4.2.

4.2 Inspection.

4.2.1 *Visual and dimensional inspection.*—Each of the sample punches and dies, taken in accordance with table V shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any punch or die in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective punches or dies in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—Each of the sample punches or dies selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.4 to verify compliance with this specification. Any sample punch or die which does not meet

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the requirements for any of these characteristics shall be rejected, and if the number of nonconforming punches or dies in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Hardness test.*

4.4.1.1 *Punches and dies.*—The punches and dies of the sample punches taken in accordance with table V shall be subjected to a hardness test to determine conformance with 3.3.

4.4.1.2 *Drivescrews.*—Drivescrews for type I of the sample punches taken by the Government inspector in accordance with table V shall be subjected to a hardness test to determine conformance with 3.6.3.

4.4.2 *Tensile-strength test.*—The drivescrews of the sample punches taken in accordance with 4.1.3.2 shall be subjected to a tensile-strength test to determine conformance with 3.7.3 for type II.

4.4.3 *Punch cutting test.*—Sample punches selected in accordance with 4.1.3.2 shall be subjected to the applicable test specified in 4.4.3.1 and 4.4.3.2.

4.4.3.1 *Type I, light duty (chassis).*—At least twelve holes shall be punched in 14-gage (0.075-inch) mild sheet steel with each sample punch.

4.4.3.2 *Type II, heavy duty (conduit).*—At least twelve holes shall be punched in 10-gage (0.135-inch) mild sheet steel with each sample punch.

4.4.3.3 Failure of the sample punches shall have occurred should the punch or die break, chip, or spall or the cutting edges become dulled to any appreciable extent, or the drivescrew or nut show any deformation.

4.5 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming punches. The Government inspector shall again examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified (see 6.1), commercial packages are acceptable.

5.2 Packing.—Unless otherwise specified (see 6.1), the subject commodity shall be delivered in a substantial container so constructed as to permit acceptance by common or other carrier for safe transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation to the point of delivery. The container shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment and shall be capable of withstanding storage, rough handling, and reshipment without the necessity of repacking.

5.3 Marking.—Unless otherwise specified (see 6.1), shipping containers shall be marked with the name of the commodity, specification number, classification, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor and the number of the contract or purchase order.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following and purchasers should exercise any desired options offered herein. (See 5.1, 5.2, 5.3.)

(a) Title, number, and date of this specification.

(b) Type and class required. (See 1.2.1.)

(c) Hole size required (see applicable table).

(d) Whether preparation for domestic shipment, immediate use; domestic shipment and storage or overseas shipment is required (see Section 5 and 7.2.3).

6.2 THE ILLUSTRATIONS HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE and are not intended to preclude the purchase of punches otherwise conforming to this specification.

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6.3 **Transportation description.**—Transportation description applicable to this item is:

Tools not otherwise indexed by name.
Carload weight 30,000 pounds.

Mechanic's hand tools not otherwise indexed.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely, related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy and Air Force.**

7.2.1 *Applicable specification and standard.*

Military Specification:

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–130—Identification Marking of U. S. Military Property.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Identification of product.*—The punches and dies shall be marked for identification in accordance with Military Standard MIL–STD–130.

7.2.3 *Preparation for delivery.*—Punches and dies shall be cleaned, preserved, unit packaged, packed, and marked in accordance with Military Specification MIL–P–15424.

MILITARY INTEREST:
Army—O Sig
Navy—Sh A Or
Air Force.

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